wfmR Academic Demo

Formal figure captions and notes

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Abstract

This demo shows the academic article defaults in `wfmR`: a title page with an abstract, an author affiliation and footnote, formal chart and table captions, smaller subtitle/notes/source blocks, and floats that stay close to their references. The examples use ggplot2 (Wickham 2016), standard principles for statistical graphics (Cleveland 1993; Tufte 2001), and built-in R datasets so that the document can be rendered without external data files.

*This is a demonstration author note for the academic PDF title page.

Descriptive Evidence

Chart 1 provides a basic descriptive view of fuel economy by cylinder count. This section is intentionally long enough to test whether floats move to the top or bottom of nearby pages without separating images, captions, notes, and sources. In an applied paper, this is the common case: a paragraph introduces the object, the object appears near the reference, and the reader can understand the chart without searching elsewhere for metadata.

The academic style differs from the report style in two important ways. First, figure captions sit below the image, while table captions sit above the table. Second, the subtitle, notes, and source are kept in a smaller details block that travels with the float. The layout is deliberately plain: the caption label is centred, the caption text is not bold, and the surrounding prose remains in the normal article text block.

The prose also uses ordinary cross-references throughout. This matters for PDF testing because the reference to Chart 1 appears before the figure itself. If LaTeX has to move the chart, the float should still land close to the first reference while preserving the title, image, subtitle, notes, and source as one grouped object. The same rule is checked later by one wide table that is large enough to make the placement decision visible.

The next paragraph is dummy placement text. It is intentionally plain and repetitive so the figure is not trivially placed where it appears in the source. The reference has already appeared, and the chart should now be eligible for the top or bottom of the nearest page depending on available space. A manuscript paragraph in this position would usually describe data construction, sample restrictions, or the intuition behind the descriptive comparison.

A second dummy paragraph lengthens the gap before the first figure. The aim is to make the contact sheet reveal whether LaTeX is managing a float queue or whether the object is being pinned inline. The grouped caption, image, subtitle, notes, and source should move together.

Bivariate Pattern

Chart 2 uses a scatter plot with a fitted line. It checks that a chart with a legend, a subtitle-style y-axis label, and a formal caption still feels coherent when it appears inside the academic PDF format. The example follows the same broad grammar-of-graphics logic as Wilkinson (2005): variables are mapped to position and colour, the fitted line is a separate layer, and the scale choices remain part of the theme rather than the substantive claim.

Dummy placement prose continues here. The next chart is referenced early in the section, but the source object appears only after this paragraph. If top/bottom floating is working, the chart should not carve out a mid-page slot simply because the code

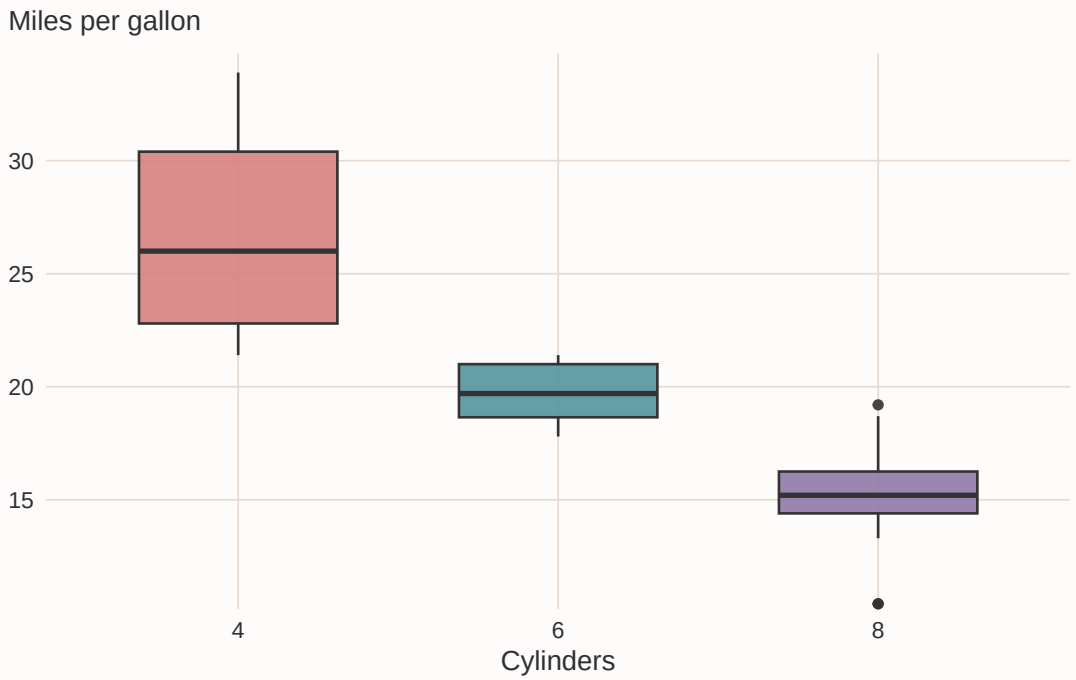


CHART 1: Fuel economy differs by cylinder count.
A simple example using the built-in mtcars data. Boxes show the interquartile range and median. Whiskers use ggplot2 defaults. R datasets package.

chunk is written here.

The additional text keeps the scatter plot from landing trivially in source order. A real empirical paper would use this space to explain identification, measurement, or modelling choices; the demo uses it to make page composition visible.

Distributional Checks

Chart 3 and Table 1 add two more object types before the palette stress test. The density chart checks overlapping semi-transparent fills and a legend, while the table is deliberately wide enough to exercise caption placement, PDF scaling, and table float behaviour in one object.

This is a useful applied-paper stress test because distributional plots often sit in the narrative before model results, and wide descriptive tables often appear as supporting evidence. The visual design should therefore stay disciplined even when several labelled objects occur within a short span of prose (Cleveland 1993).

Dummy placement text makes this section behave more like a manuscript page than a gallery. The chart has already been referenced, but the source chunk is still below this paragraph. LaTeX should be free to decide whether the density chart belongs at the bottom of the current page or at the top of the next.

The wide table is deliberately introduced in the same narrative block. That creates pressure on the float queue and makes it easier to see whether tables are genuine

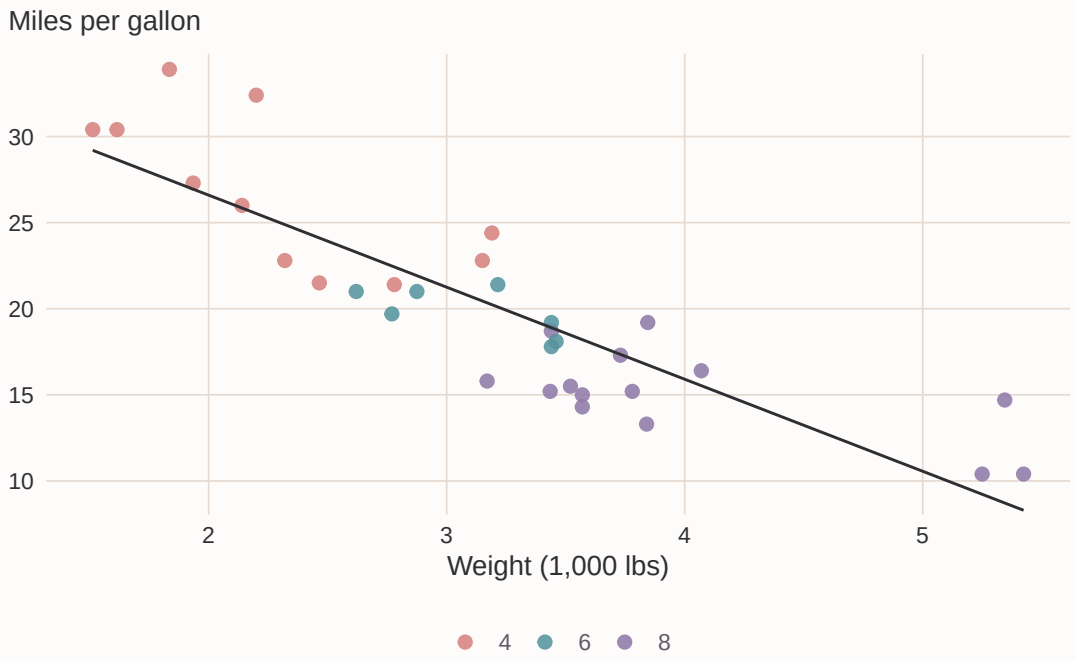


CHART 2: Fuel economy declines with vehicle weight.
Points are grouped by cylinder count. The fitted line is an ordinary least squares fit shown for visual orientation. R datasets package.

floats. If the table appears mid-page because it is a `longtable`, the test has failed.

Table 1 should inherit the formal table caption treatment, with the caption above the table and the small details below. This is also a quick check that shaded HTML rows remain subtle while the PDF table keeps the plain academic appearance.

Time-Series Example

Chart 4 uses the built-in `AirPassengers` series to test a different data shape. The figure is a line chart rather than a categorical comparison, which gives the PDF output another opportunity to place a figure at the top or bottom of a nearby page while preserving the figure's caption group.

The time-series example also checks that a single salmon-blue accent chart does not look out of place next to the more categorical palette examples. That is the kind of comparison a template needs to survive in ordinary manuscript drafting, where figures rarely all share the same geometry.

This dummy paragraph gives the time-series chart a realistic delay after its reference. A chart inserted in source order should still float to a page edge when the text block is already partly filled. The point is not the data; the point is the page behaviour.

The extra prose is intentionally separated from the chart chunk. This keeps the remaining pages realistic without adding more table examples.

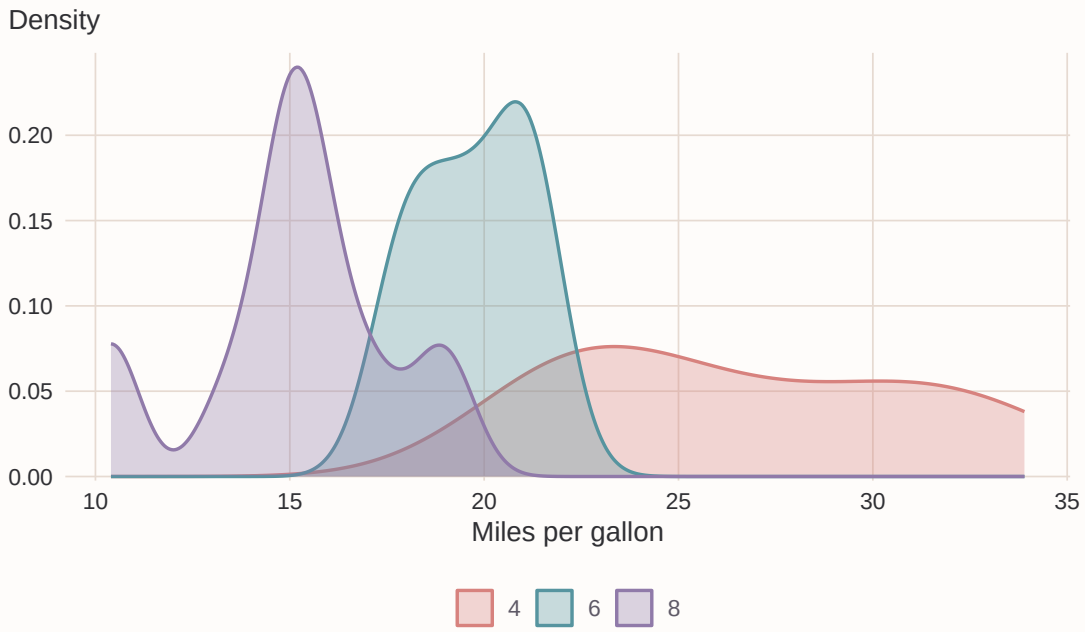


CHART 3: Fuel-economy distributions differ by cylinder count.
Density curves provide a crowded legend example. The curves are illustrative and use the small built-in mtcars data. R datasets package.

Palette Checks

Chart 5 is synthetic. It uses all seven colours in the default palette order so that a single page can reveal whether the salmon, teal, purple, lime, yellow, blue, and pink sequence remains legible in the formal academic setting. The choice is intentionally softer than a strict publication-first palette, but the example keeps the sequence explicit for brand consistency and for comparison with colour-documentation guidance such as Brewer, Hatchard, and Harrower (2003).

The final chart is again separated from its reference by dummy text. It should therefore make clear whether the remaining material is being handled as a float queue. The bibliography barrier at the end of the document is retained only to prevent a late float from drifting into the references.

TABLE 1: Wide vehicle summary.

Toyota Corona	4	3	Automatic	21.5	2.46	97	20.01	3.70	120
Datsun 710	4	4	Manual	22.8	2.32	93	18.61	3.85	108
Fiat 128	4	4	Manual	32.4	2.20	66	19.47	4.08	79
Fiat X1-9	4	4	Manual	27.3	1.94	66	18.90	4.08	79
Honda Civic	4	4	Manual	30.4	1.61	52	18.52	4.93	76
Merc 230	4	4	Automatic	22.8	3.15	95	22.90	3.92	141
Merc 240D	4	4	Automatic	24.4	3.19	62	20.00	3.69	147
Toyota Corolla	4	4	Manual	33.9	1.83	65	19.90	4.22	71
Volvo 142E	4	4	Manual	21.4	2.78	109	18.60	4.11	121
Lotus Europa	4	5	Manual	30.4	1.51	113	16.90	3.77	95
Porsche 914-2	4	5	Manual	26.0	2.14	91	16.70	4.43	120
Hornet 4 Drive	6	3	Automatic	21.4	3.21	110	19.44	3.08	258
Valiant	6	3	Automatic	18.1	3.46	105	20.22	2.76	225
Mazda RX4	6	4	Manual	21.0	2.62	110	16.46	3.90	160
Mazda RX4 Wag	6	4	Manual	21.0	2.88	110	17.02	3.90	160
Merc 280	6	4	Automatic	19.2	3.44	123	18.30	3.92	168
Merc 280C	6	4	Automatic	17.8	3.44	123	18.90	3.92	168
Ferrari Dino	6	5	Manual	19.7	2.77	175	15.50	3.62	145
AMC Javelin	8	3	Automatic	15.2	3.44	150	17.30	3.15	304
Cadillac Fleetwood	8	3	Automatic	10.4	5.25	205	17.98	2.93	472
Camaro Z28	8	3	Automatic	13.3	3.84	245	15.41	3.73	350
Chrysler Imperial	8	3	Automatic	14.7	5.34	230	17.42	3.23	440
Dodge Challenger	8	3	Automatic	15.5	3.52	150	16.87	2.76	318
Duster 360	8	3	Automatic	14.3	3.57	245	15.84	3.21	360
Hornet Sportabout	8	3	Automatic	18.7	3.44	175	17.02	3.15	360
Lincoln Continental	8	3	Automatic	10.4	5.42	215	17.82	3.00	460
Merc 450SE	8	3	Automatic	16.4	4.07	180	17.40	3.07	276
Merc 450SL	8	3	Automatic	17.3	3.73	180	17.60	3.07	276
Merc 450SLC	8	3	Automatic	15.2	3.78	180	18.00	3.07	276
Pontiac Firebird	8	3	Automatic	19.2	3.85	175	17.05	3.08	400
Ford Pantera L	8	5	Manual	15.8	3.17	264	14.50	4.22	351
Maserati Bora	8	5	Manual	15.0	3.57	335	14.60	3.54	301

Selected car-level measures from the built-in mtcars data. The table is intentionally wide and uses all 32 cars so that PDF placement and scaling remain visible. R datasets package.

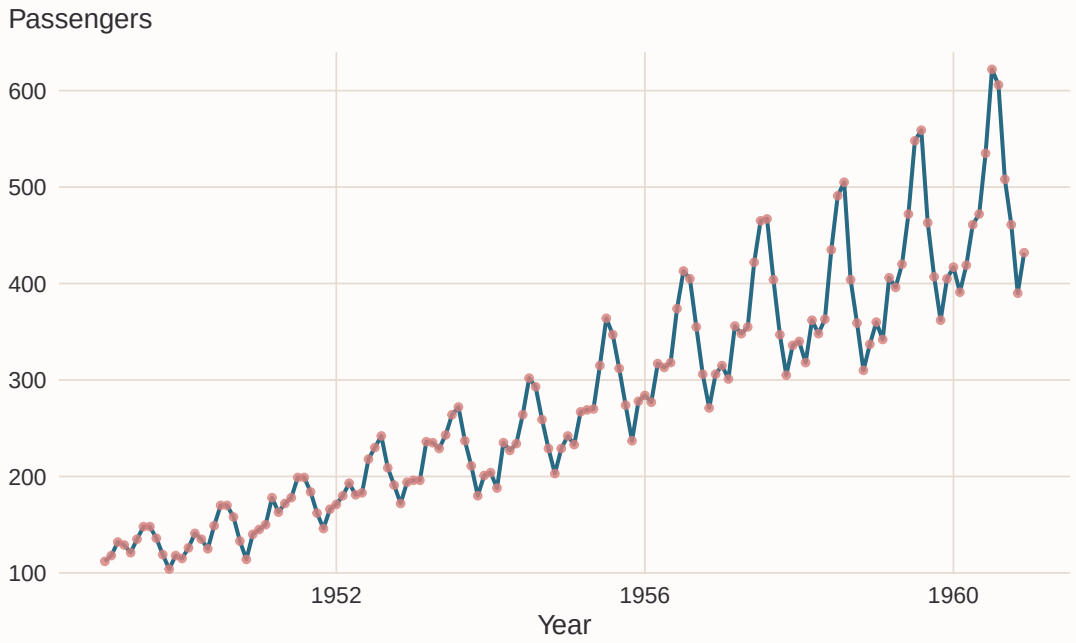


CHART 4: Passenger counts rise sharply in the late 1950s.

Monthly totals from the built-in AirPassengers series. The example is included to exercise a time-series chart in the same academic style. R datasets package.

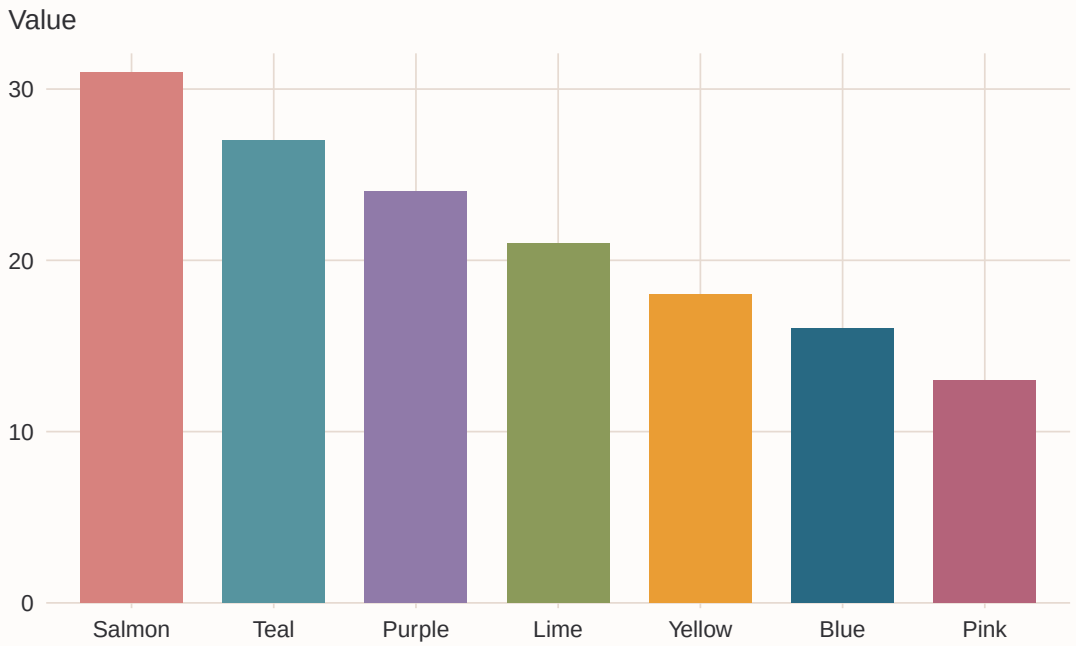


CHART 5: The academic style keeps the wfmR palette order.

Seven bars exercise every default categorical colour. The values are illustrative and are ordered only to show the default colour sequence. Illustrative data.

References

Brewer, Cynthia A., Geoffrey W. Hatchard, and Mark A. Harrower. 2003. "ColorBrewer in Print: A Catalog of Color Schemes for Maps." *Cartography and Geographic Information Science* 30 (1): 5–32. https://doi.org/10.1559/152304003100010929.

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